

30/11/19
 Q. Trace Quick sort for the following array.

3	8	12	19	31	46	54
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54	12	19	3	8	46	31
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$i > j$

Swap i & pivot

31	12	19	3	8	46	54
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qs(a, 0, 5) ✓

qs(a, 6, 6) ✗

31	12	19	8	31	46	54
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$i > j \Rightarrow$ swap i & pivot

qs(a, 0, 3) ✓

qs(a, 4, 5) ✓

$n = 7$; first = 0, last = 6

i j $a[i]$ $a[j]$ $a[\text{pivot}]$

0 6 54 31 54

initial

(swap i & pivot j)

- 6 - 31 54

So,

3	12	19	8	31	46	54
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3	12	19	8	31	46	54
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$i > j \Rightarrow$ swap pivot j

i j j j

3	12	19	8	31	46	54
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i i j

swaps: (3 | 8 | 12 | 19 | 31 | 46 | 84)

i	j	a[i]	a[j]	a[pivot]	
0	5	31	46	31	initial
3	4	3	8	31	$i < j \Rightarrow \text{swap } i \& j$
0	5	31	46	31	
5	4	46	3	31	$i > j \Rightarrow \text{swap pivot \& j}$
0	3	3	8	3	initial
1	0	12	3	3	swap j \& pivot
1	3	12	8	12	initial
2	3	19	8	12	swap i \& j
1	3	12	19	12	initial
3	2	19	8	12	swap pivot \& j

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